

1 Architectural contribution

A 0.8B-parameter CPU classifier replaces the LLM consultant. Routing runs 30× faster per turn (~50 ms vs. ~1.5 s API round-trip) and eliminates the 21% JSON-schema fallback rate at scale. State accuracy lifts 2.14× (+29.45 pp) and the unified score rises +19.25 points over the GPT-4o + SocratTeachLLM paper baseline. All at \$0 API cost on one 32 GB consumer GPU.

2 Methodological contribution

Five converging diagnostics establish that ROUGE/BLEU on this benchmark reward training-data memorization, not transferable teaching capability. Three are shown above (cross-lingual translation, clean-probe collapse, within-architecture ablation); two more (rank inversion and n-gram-length scaling) are detailed in the paper.

3 Limitations and future work

Result is specific to this teacher, prompt, and consultant (Chinese-language SocratDataset). The +2.18 frontier-overtaking margin is supported but bounded by the n=400 variance budget. A native English-domain benchmark and bilingual classifier co-training are next.

4 Real classrooms, today

Elementary-school classrooms anywhere are within reach today. The full pipeline fits on a single consumer-grade GPU at near-zero per-run cost, and hardware like Nvidia's newly released N1 chip enables deployment on shared classroom rigs and individual student laptops alike. Completing the scaling work to every language and subject is how every student gets access, not only those at well-funded schools.